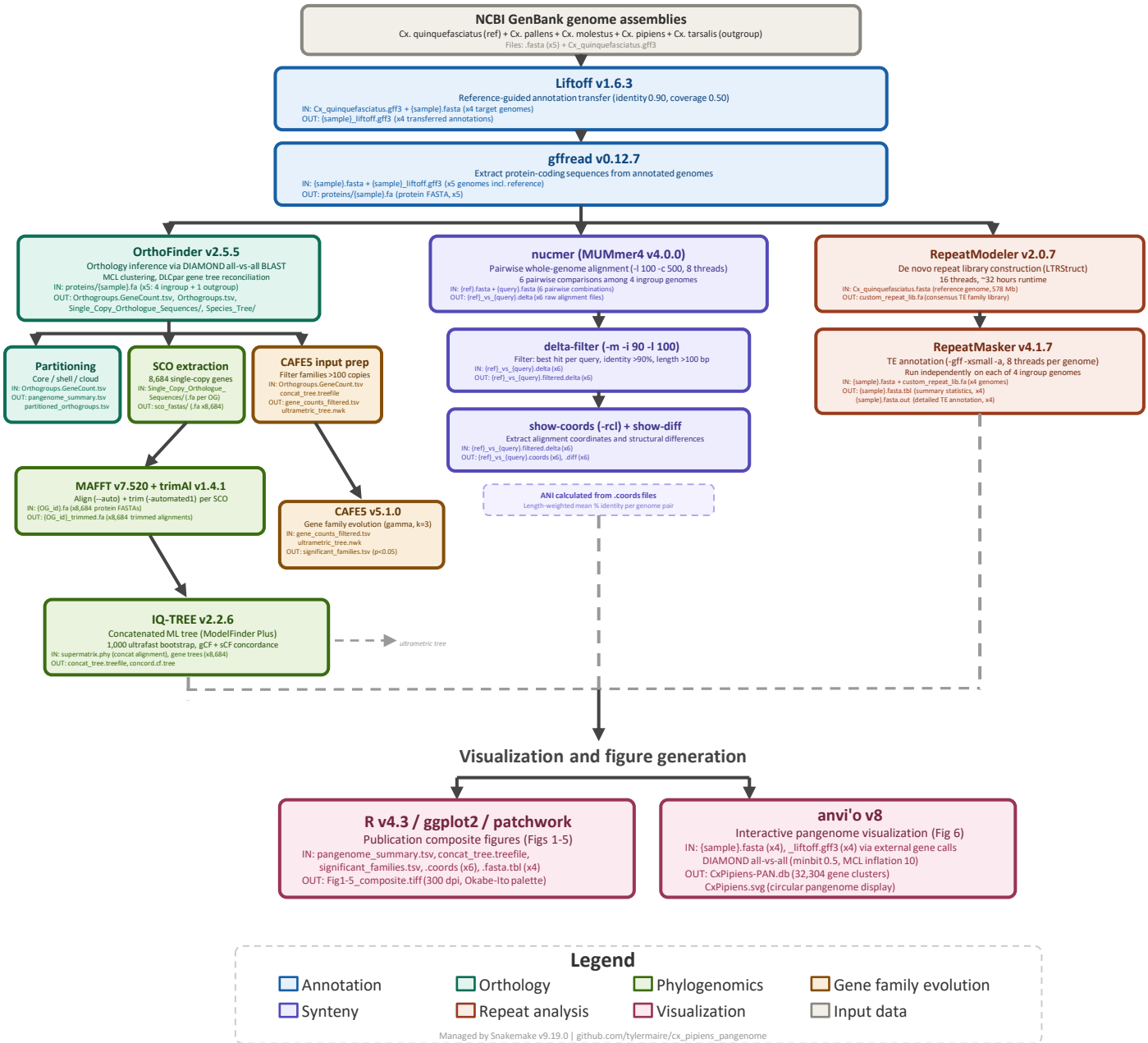
NCBI GenBank genome assemblies
Cx. quinquefasciatus (ref) + Cx. pallens + Cx. molestus + Cx. pipiens + Cx. tarsalis (outgroup)
Files: .fasta (x5) + Cx_quinquefasciatus.gff3
Liftoff v1.6.3
Reference-guided annotation transfer (identity 0.90, coverage 0.50)
IN: Cx_quinquefasciatus.gff3 + {sample}.fasta (x4 target genomes)
OUT: {sample}_liftoff.gff3 (x4 transferred annotations)
gffread v0.12.7
Extract protein-coding sequences from annotated genomes
IN: {sample}.fasta + {sample}_liftoff.gff3 (x5 genomes incl. reference)
OUT: proteins/{sample}.fa (protein FASTA, x5)
OrthoFinder v2.5.5
Orthology inference via DIAMOND all-vs-all BLAST
MCL clustering, DLCpar gene tree reconciliation
IN: proteins/{sample}.fa (x5: 4 ingroup + 1 outgroup)
OUT: Orthogroups.GeneCount.tsv, Orthogroups.tsv,
Single_Copy_Orthologue_Sequences/, Species_Tree/
nucmer (MUMmer4 v4.0.0)
Pairwise whole-genome alignment (-l 100 -c 500, 8 threads)
6 pairwise comparisons among 4 ingroup genomes
IN: {ref}.fasta + {query}.fasta (6 pairwise combinations)
OUT: {ref}_vs_{query}.delta (x6 raw alignment files)
RepeatModeler v2.0.7
De novo repeat library construction (LTRStruct)
16 threads, ~32 hours runtime
IN: Cx_quinquefasciatus.fasta (reference genome, 578 Mb)
OUT: custom_repeat_lib.fa (consensus TE family library)
Partitioning
Core / shell / cloud
IN: Orthogroups.GeneCount.tsv
OUT: pangenome_summary.tsv
partitioned_orthogroups.tsv
SCO extraction
8,684 single-copy genes
IN: Single_Copy_Orthologue_
Sequences/ (.fa per OG)
OUT: sco_fastas/ (.fa x8,684)
CAFE5 input prep
Filter families >100 copies
IN: Orthogroups.GeneCount.tsv
concat_tree.treefile
OUT: gene_counts_filtered.tsv
ultrametric_tree.nwk
delta-filter (-m -i 90 -l 100)
Filter: best hit per query, identity >90%, length >100 bp
IN: {ref}_vs_{query}.delta (x6)
OUT: {ref}_vs_{query}.filtered.delta (x6)
RepeatMasker v4.1.7
TE annotation (-gff -xsmall -a, 8 threads per genome)
Run independently on each of 4 ingroup genomes
IN: {sample}.fasta + custom_repeat_lib.fa (x4 genomes)
OUT: {sample}.fasta.tbl (summary statistics, x4)
{sample}.fasta.out (detailed TE annotation, x4)
show-coords (-rcl) + show-diff
Extract alignment coordinates and structural differences
IN: {ref}_vs_{query}.filtered.delta (x6)
OUT: {ref}_vs_{query}.coords (x6), .diff (x6)
ANI calculated from .coords files
Length-weighted mean % identity per genome pair
MAFFT v7.520 + trimAl v1.4.1
Align (--auto) + trim (-automated1) per SCO
IN: {OG_id}.fa (x8,684 protein FASTAs)
OUT: {OG_id}_trimmed.fa (x8,684 trimmed alignments)
CAFE5 v5.1.0
Gene family evolution (gamma, k=3)
IN: gene_counts_filtered.tsv
ultrametric_tree.nwk
OUT: significant_families.tsv (p<0.05)
IQ-TREE v2.2.6
Concatenated ML tree (ModelFinder Plus)
1,000 ultrafast bootstrap, gCF + sCF concordance
IN: supermatrix.phy (concat alignment), gene trees (x8,684)
OUT: concat_tree.treefile, concord.cf.tree
ultrametric tree
Visualization and figure generation
R v4.3 / ggplot2 / patchwork
Publication composite figures (Figs 1-5)
IN: pangenome_summary.tsv, concat_tree.treefile,
significant_families.tsv, .coords (x6), .fasta.tbl (x4)
OUT: Fig1-5_composite.tiff (300 dpi, Okabe-Ito palette)
anvi'o v8
Interactive pangenome visualization (Fig 6)
IN: {sample}.fasta (x4), _liftoff.gff3 (x4) via external gene calls
DIAMOND all-vs-all (minbit 0.5, MCL inflation 10)
OUT: CxPipiens-PAN.db (32,304 gene clusters)
CxPipiens.svg (circular pangenome display)
Legend
Annotation
Orthology
Phylogenomics
Gene family evolution
Synteny
Repeat analysis
Visualization
Input data
Managed by Snakemake v9.19.0 | github.com/tylermaire/cx_pipiens_pangenome